

Sailab Banik

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SUMMARY

Software Engineer with 3+ years of experience building production-grade full-stack and AI-powered applications, specializing in **Python, React, FastAPI, and AI/LLM systems**. Experienced in **RAG pipelines, vision-language models, document intelligence**, and end-to-end feature ownership across architecture, implementation, CI/CD, and production delivery.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), SQL, HTML, CSS
AI/ML: Generative AI, RAG Pipelines, Large Language Models (LLMs), Vision-Language Models (VLMs), LangChain, Ollama, OpenAI API, Hugging Face, Vector Databases (Chroma)
Backend: FastAPI, REST APIs, PostgreSQL, Redis, SQLAlchemy, Pydantic, OAuth2 / SSO (Microsoft Entra ID, MSAL), JWT Authentication
Frontend: React.js, Next.js, Redux Toolkit, Material UI, Vite, React Testing Library, Vitest, MSW
DevOps & Tools: Docker, Docker Compose, CI/CD Pipelines, Azure DevOps, Git, GitLab

WORK EXPERIENCE

- Software Engineer II**
ZF Group
- Oct 2024 – Present
Chennai, India
- Led frontend development of **PrimeAI**, an internal AI-powered document processing platform adopted by **5+ business teams**, reducing manual document review effort by **60%** and saving an estimated **2,000+ operational hours per quarter**.
 - Owned key end-to-end product flows across UI, API, and database layers, including AI model creation, annotation-driven extraction, and workflow automation.
 - Designed and implemented **role- and group-based access control** from the ground up, enabling secure multi-team adoption and supporting rollout to departments with strict data-access requirements.
 - Drove CI/CD improvements across the team, reducing release cycle time by **40%** and production hotfix incidents by **35%**; introduced frontend test suites using **React Testing Library and Vitest**, reducing manual testing effort by **60%**.
- Graduate Engineer Trainee**
ZF Group
- Aug 2023 – Sep 2024
Chennai, India
- Stabilized a business-critical legacy platform by resolving **10+ high-priority production bugs**, reducing user-reported issues by **45%**.
 - Remediated **1,000+ SQL injection findings** across internal applications, contributing to successful security audits and reducing application security risk.
 - Optimized operations scheduling workflows, improving throughput by **30%** and earning the **Budding Star Award** for measurable business contribution within the first year.

PROJECTS

- QueueLens: Distributed Job Queue with Live Dashboard**
Python, FastAPI, Redis, PostgreSQL, Next.js, Docker
- Built a producer-consumer job queue: a **FastAPI** producer persists jobs in **PostgreSQL** and hands job IDs to background workers over a **Redis** list (LPUSH/BRPOP), with a **Next.js** dashboard tracing every job through its lifecycle live.
 - Implemented **retries with exponential backoff**, Redis-backed **per-client rate limiting** that holds across API processes, and a queue cap returning 429/503 with Retry-After; ordered state commits before enqueueing to prevent API-worker race conditions. Shipped with Docker Compose and pytest suites on fakedis.
- Vision-Model-Powered Extraction for Complex Documents**
Python, VLMs, PP-Structure, Ollama, FastAPI
- Built an end-to-end extraction system for visually complex documents, including forms, multi-column layouts, and mixed tables/text, combining **PP-Structure** layout analysis with a locally served open-source **Vision-Language Model**.
 - Segmented pages into semantic regions and routed each region to the VLM using region-specific prompts, improving structured-field accuracy by **~25%** over raw page images; enforced **JSON-schema-guided outputs** with post-generation validation for **100% schema-valid responses**.
- Advanced RAG Pipeline for 100+ Page Documents**
Python, FastAPI, LangChain, Ollama, Qwen3, Chroma
- Built a pipeline converting unstructured **100+ page PDFs** into structured JSON from user-defined schemas, running fully on local infrastructure with **Ollama-hosted Qwen3** and keeping sensitive data within the environment.
 - Designed a **dual-path RAG architecture** with separate text and table vector stores, structure-aware chunking, and cross-encoder reranking, improving retrieval precision by **~40%** over a single-store baseline.
 - Implemented runtime validation using **dynamically generated Pydantic models** and an **LLM-driven repair loop**, improving valid-JSON responses from **~75% to 100%**; exposed the pipeline through a modular REST API with OpenAPI documentation and idempotent ingestion.

EDUCATION

- ICFAI University Tripura**
B.Tech in Computer Science and Engineering – **CGPA: 9.64**
- Aug 2019 – Jul 2023
Agartala, India

CERTIFICATIONS & AWARDS

- Budding Star Award** — ZF Group, Recognition for Business Impact
System Design Cohort — ChaiCode: distributed systems, load balancing, caching, sharding, queues
AI Engineer Core Track (LLM Engineer) — Udemy: production LLM apps, RAG, agentic workflows, fine-tuning
- 2024
[Credential](#) | Jun 2026
[Credential](#) | May 2026